

Curvature-Aware Zero-Offset Hyperbolic Regularization (C-Z0HR): Mathematical Proofs, C^∞ Smooth-Norm Derivations, and Multi-Campaign Benchmark Audits

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Abstract

Spurious turbulence quenching and unphysical surface cold biases in numerical weather prediction (NWP) models stem from steep empirical stability closures interacting with under-resolved coordinate curvature near critical Richardson thresholds ($Ri_c \approx 0.20\text{--}0.25$) (Nieuwstadt, 1984; Louis, 1979). This work establishes the formal mathematical foundation of Curvature-Aware Zero-Offset Hyperbolic Regularization (C-Z0HR). By shifting the regularization operator upstream directly into the input gradient Richardson number ($\widetilde{Ri}_g$), C-Z0HR preserves legacy stability functions ($f_{m,h}$) without empirical parameter retuning (England and McNider, 1995). We present rigorous proofs of threshold fixed-point invariance, distance contraction, and partial derivative slope attenuation, introduce a C^∞ smooth-norm curvature operator, derive an end-to-end differentiable regularized ramp closure, and evaluate benchmark audits across CASES-99, GABLS3, and SHEBA field campaign datasets (Poulos et al., 2002; Bosveld et al., 2014; Uttal et al., 2002).

1 Curvature-Aware Zero-Offset Hyperbolic Regularization (C-Z0HR) Proofs

1.1 Upstream Operator Formulation

Standard boundary-layer parameterizations modify downstream exchange coefficients $K_{m,h}$ or empirical stability functions $f_{m,h}(Ri)$ (Businger et al., 1971; Louis, 1979). C-Z0HR instead regularizes the diagnostic scalar argument Ri_g upstream before closure evaluation.

Let $Ri_g(z)$ be the local gradient Richardson number, Ri_c the critical Richardson threshold ($Ri_c \approx 0.20$), $\alpha > 0$ the damping control parameter (standard $\alpha = 2.0$), and $\epsilon_c Ri_c$ a non-zero softening floor ($\epsilon_c \approx 0.05$). The local vertical wind shear magnitude is denoted by the calligraphic operator $\mathcal{S} \equiv \left| \frac{\partial \mathbf{u}}{\partial z} \right| = \sqrt{\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2}$.

To guarantee C^∞ differentiability of the spatial operator across inflection points where $\frac{\partial^2 Ri_g}{\partial z^2} = 0$, local spatial curvature $C_\eta(z, \Delta z)$ is evaluated using a smooth C^∞ norm $|q|_\eta \equiv \sqrt{q^2 + \eta^2}$ with $\eta = 10^{-4}$:

$$C_\eta(z, \Delta z) = \frac{1}{2} \sqrt{\left(\frac{\partial^2 Ri_g}{\partial z^2} \right)^2 + \eta^2 (\Delta z)^2 + C_{\text{floor}}}, \quad C_{\text{floor}} \equiv \epsilon_c Ri_c \quad (C_\eta > 0). \quad (1)$$

To eliminate non-differentiable C^0 kinks ($|Ri_g - Ri_c|$) at threshold crossing, C-Z0HR introduces the real-analytic (C^∞) hyperbolic metric (Tikhonov, 1963):

$$\Phi_\epsilon(x) = \sqrt{x^2 + \epsilon^2}, \quad x \equiv Ri_g - Ri_c, \quad (2)$$

where $\epsilon \approx 10^{-3}$ parameterizes smooth metric transition width. The regularized Richardson number mapping $\widetilde{Ri}_g$ is defined as:

$$\widetilde{Ri}_g = Ri_c + (Ri_g - Ri_c) \left[\frac{\Phi_\epsilon(Ri_g - Ri_c) + C_\eta}{\Phi_\epsilon(Ri_g - Ri_c) + (1 + \alpha)C_\eta} \right]. \quad (3)$$

The regularized scalar $\widetilde{Ri}_g$ feeds directly into unmodified legacy exchange functions:

$$K_m = l_m^2 \mathcal{S} \cdot f_m(\widetilde{Ri}_g), \quad K_h = l_h^2 \mathcal{S} \cdot f_h(\widetilde{Ri}_g). \quad (4)$$

1.2 Fundamental Properties of the Mapping

Proposition 1 (Threshold Fixed-Point Invariance / Zero-Offset). *For any $C_\eta > 0$ and $\alpha > 0$, the critical threshold Ri_c is an exact fixed point of the regularized mapping:*

$$Ri_g = Ri_c \iff \widetilde{Ri}_g = Ri_c. \quad (5)$$

Thus, C-Z0HR is "zero-offset": it compresses distance from criticality without shifting the critical physical threshold Ri_c .

Proof. Setting $Ri_g = Ri_c \implies x = 0$. Substituting $x = 0$ into eq. (3) gives $\widetilde{Ri}_g = Ri_c + 0 \cdot M(0, C_\eta) = Ri_c$. Conversely, if $\widetilde{Ri}_g = Ri_c$, then $x \cdot M(x, C_\eta) = 0$. Since $M(x, C_\eta) = \frac{\Phi_\epsilon + C_\eta}{\Phi_\epsilon + (1 + \alpha)C_\eta} > 0$ for all x , it follows that $x = 0 \implies Ri_g = Ri_c$. $\square$

Proposition 2 (Distance Contraction and Sign Preservation). *For finite $C_\eta > 0$ and $\alpha > 0$, the mapping preserves stability sign relative to Ri_c and strictly compresses distance to criticality:*

$$\text{sgn}(\widetilde{Ri}_g - Ri_c) = \text{sgn}(Ri_g - Ri_c), \quad 0 < \frac{|\widetilde{Ri}_g - Ri_c|}{|Ri_g - Ri_c|} < 1 \quad (x \neq 0). \quad (6)$$

Proof. Let $M(x, C_\eta) \equiv \frac{\Phi_\epsilon(x) + C_\eta}{\Phi_\epsilon(x) + (1 + \alpha)C_\eta}$. Since $\alpha > 0$ and $C_\eta > 0$, we have $0 < \Phi_\epsilon + C_\eta < \Phi_\epsilon + (1 + \alpha)C_\eta$, establishing $0 < M(x, C_\eta) < 1$. Multiplying by $x = Ri_g - Ri_c$ yields $\widetilde{Ri}_g - Ri_c = x \cdot M(x, C_\eta)$, preserving sign while strictly contracting magnitude. $\square$

1.3 Partial Slope and Attenuation Proof

Theorem 1 (Frozen-Curvature Partial Mapping Slope). *Holding spatial curvature $C_\eta > 0$ fixed ($\partial C_\eta / \partial Ri_g \approx 0$), the partial mapping slope $\mathcal{D}_\alpha^{(R)}(x) \equiv \left. \frac{\partial \widetilde{Ri}_g}{\partial Ri_g} \right|_{C_\eta}$ is exact for all $x \in \mathbb{R}$:*

$$\mathcal{D}_\alpha^{(R)}(x) = \frac{\Phi_\epsilon(x) + C_\eta}{\Phi_\epsilon(x) + (1 + \alpha)C_\eta} + \frac{\alpha C_\eta x^2}{\Phi_\epsilon(x) [\Phi_\epsilon(x) + (1 + \alpha)C_\eta]^2}. \quad (7)$$

Proof. Let $x \equiv Ri_g - Ri_c$ and write eq. (3) as $\widetilde{Ri}_g = Ri_c + x \frac{N(x)}{D(x)}$, where $N(x) = \Phi_\epsilon(x) + C_\eta$ and $D(x) = \Phi_\epsilon(x) + (1 + \alpha)C_\eta$. Differentiating with respect to x at fixed C_η :

$$\mathcal{D}_\alpha^{(R)} = \frac{\partial \widetilde{Ri}_g}{\partial x} = \frac{N(x)}{D(x)} + x \frac{N'(x)D(x) - N(x)D'(x)}{[D(x)]^2}. \quad (8)$$

Since $\Phi'_\epsilon(x) = \frac{x}{\sqrt{x^2 + \epsilon^2}} = \frac{x}{\Phi_\epsilon(x)}$, we have $N'(x) = D'(x) = \frac{x}{\Phi_\epsilon(x)}$. Substituting these derivatives yields:

$$N'(x)D(x) - N(x)D'(x) = \frac{x}{\Phi_\epsilon(x)} \left[(\Phi_\epsilon + (1 + \alpha)C_\eta) - (\Phi_\epsilon + C_\eta) \right] = \frac{\alpha C_\eta x}{\Phi_\epsilon(x)}. \quad (9)$$

Substituting back gives the exact partial slope in eq. (7). $\square$

Corollary 1 (Critical-Point Slope Attenuation Factor). *At exact threshold crossing ($x = 0 \implies \Phi_\epsilon(0) = \epsilon$), the x^2 term in eq. (7) vanishes identically, yielding the exact critical slope:*

$$\mathcal{D}_\alpha^{(R)}(0) = \frac{\epsilon + C_\eta}{\epsilon + (1 + \alpha)C_\eta}. \quad (10)$$

In the limit as $\epsilon/C_\eta \rightarrow 0$, this converges to the limiting critical-point slope attenuation factor:

$$\lim_{\epsilon/C_\eta \rightarrow 0} \mathcal{D}_\alpha^{(R)}(0) = \frac{1}{1 + \alpha}. \quad (11)$$

For standard damping $\alpha = 2.0$, $\lim_{\epsilon/C_\eta \rightarrow 0} \mathcal{D}_2^{(R)}(0) = 1/3$, establishing a 66.7% reduction in critical slope sensitivity.

Proposition 3 (Full Spatial Jacobian Decomposition). *When local curvature $C_\eta(Ri_g, z, \Delta z)$ is updated dynamically, the total spatial derivative of the regularized field with respect to raw Ri_g decomposes into:*

$$\frac{d\widetilde{Ri}_g}{dRi_g} = \mathcal{D}_\alpha^{(R)} + \frac{\partial \widetilde{Ri}_g}{\partial C_\eta} \frac{dC_\eta}{dRi_g}, \quad (12)$$

where $\mathcal{D}_\alpha^{(R)}$ isolates frozen-curvature threshold attenuation, while $\frac{\partial \widetilde{Ri}_g}{\partial C_\eta} \frac{dC_\eta}{dRi_g}$ accounts for non-local vertical grid stencil coupling.

1.4 Local Thermal Feedback Sensitivity

In Single-Column Models (SCMs), evaluating local discrete sensitivity across the thermal coupling chain under frozen curvature ($\partial C_\eta / \partial Ri_g \approx 0$) yields:

$$\mathcal{G}_{\text{reg}} = \underbrace{\left(\frac{\partial Ri_g}{\partial \theta_z} \right) \left(\frac{\partial \theta_z}{\partial H} \right) \left(\frac{\partial H}{\partial K_h} \right) K'_h(\widetilde{Ri}_g)}_{\mathcal{G}_{\text{physical}}} \cdot \mathcal{D}_\alpha^{(R)}(Ri_g). \quad (13)$$

Enforcing $\mathcal{D}_\alpha^{(R)}(Ri_c) \approx \frac{1}{1+\alpha}$ provides a sufficient local condition for discrete contraction ($|\mathcal{G}_{\text{reg}}(Ri_c)| < 1$):

$$|\mathcal{G}_{\text{physical}}(Ri_c)| < 1 + \alpha \implies |\mathcal{G}_{\text{reg}}(Ri_c)| < 1. \quad (14)$$

For $\alpha = 2.0$, local sensitivity amplification is contained whenever $|\mathcal{G}_{\text{physical}}| < 3$. Full numerical stability in 3D NWP solvers depends on time-step size, spatial advection discretization, and off-diagonal stencil terms ($\partial C_\eta / \partial Ri_g$).

2 C^∞ Regularized Ramp Closure & Analytical Jacobian Derivations

2.1 Regularized Ramp Operator Formulation

To guarantee C^∞ differentiability when replacing non-smooth threshold ramps ($(g)_+ = \max(0, g)$), C-Z0HR employs the infinitely differentiable regularized ramp operator $\mathcal{R}_\epsilon$:

$$\mathcal{R}_\epsilon(g) \equiv \frac{1}{2} \left(g + \sqrt{g^2 + \epsilon^2} \right), \quad g \equiv 1 - \frac{\widetilde{Ri}_g}{Ri_c}. \quad (15)$$

The regularized momentum stability function f_m is formulated as:

$$f_m(\widetilde{Ri}_g) = \left[\mathcal{R}_\epsilon \left(1 - \frac{\widetilde{Ri}_g}{Ri_c} \right) \right]^2. \quad (16)$$

2.2 Analytical Jacobian Matrix Derivation

Theorem 2 (Smooth Threshold Closure & Jacobian). *For $\epsilon > 0$, $f_m \in C^\infty(\mathbb{R})$ with respect to $\widetilde{Ri}_g$. Under fixed $C_\eta > 0$, the partial derivative with respect to input Ri_g exists and is C^∞ -continuous everywhere:*

$$\frac{\partial f_m}{\partial Ri_g} = -\frac{\mathcal{R}_\epsilon(g)}{Ri_c} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right) \mathcal{D}_\alpha^{(R)}(Ri_g). \quad (17)$$

At exact criticality ($\widetilde{Ri}_g = Ri_c \implies g = 0$), the local derivative evaluates to:

$$\left. \frac{\partial f_m}{\partial \widetilde{Ri}_g} \right|_{\widetilde{Ri}_g = Ri_c} = -\frac{\epsilon}{2Ri_c}. \quad (18)$$

Proof. Applying the chain rule: $\frac{\partial f_m}{\partial Ri_g} = \frac{\partial f_m}{\partial \widetilde{Ri}_g} \cdot \frac{\partial \widetilde{Ri}_g}{\partial Ri_g}$. Differentiating eq. (16) with respect to $\widetilde{Ri}_g$:

$$\frac{\partial f_m}{\partial \widetilde{Ri}_g} = 2\mathcal{R}_\epsilon(g) \cdot \frac{\partial \mathcal{R}_\epsilon(g)}{\partial g} \cdot \frac{\partial g}{\partial \widetilde{Ri}_g}. \quad (19)$$

From $\frac{\partial g}{\partial \widetilde{Ri}_g} = -\frac{1}{Ri_c}$ and $\frac{\partial \mathcal{R}_\epsilon(g)}{\partial g} = \frac{1}{2} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right)$:

$$\frac{\partial f_m}{\partial \widetilde{Ri}_g} = 2\mathcal{R}_\epsilon(g) \cdot \frac{1}{2} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right) \left(-\frac{1}{Ri_c} \right) = -\frac{\mathcal{R}_\epsilon(g)}{Ri_c} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right). \quad (20)$$

At $g = 0$, $\mathcal{R}_\epsilon(0) = \epsilon/2$ and $(1 + \frac{0}{\epsilon}) = 1$, yielding $\left. \frac{\partial f_m}{\partial \widetilde{Ri}_g} \right|_{g=0} = -\frac{\epsilon}{2Ri_c}$. Multiplying by $\mathcal{D}_\alpha^{(R)}(Ri_g)$ completes the proof. $\square$

2.3 Supercritical Asymptotic Tail

Proposition 4 (Supercritical Tail and Strict Positivity). *Under strongly stable conditions ($\widetilde{Ri}_g \gg Ri_c \implies g < 0, |g| \gg \epsilon$), using $g + \sqrt{g^2 + \epsilon^2} = \frac{\epsilon^2}{\sqrt{g^2 + \epsilon^2} - g} \approx \frac{\epsilon^2}{2|g|}$, the closure exhibits the asymptotic tail:*

$$f_m(\widetilde{Ri}_g) \approx \left(\frac{\epsilon^2}{4|g|} \right)^2 = \frac{\epsilon^4}{16 \left(\frac{\widetilde{Ri}_g}{Ri_c} - 1 \right)^2} > 0. \quad (21)$$

For finite $\widetilde{Ri}_g$ and $\epsilon > 0$, $f_m > 0$ strictly, preserving strict positivity of $K_m > 0$ and ensuring continuous Jacobians ($\partial K_m / \partial \theta_z$) for implicit backward-Euler time-steppers.

3 Multi-Campaign Benchmark Matrix & Empirical Audits

To evaluate C-Z0HR across diverse stability regimes, diagnostic audits were conducted on field campaign profiles from CASES-99 (Poulos et al., 2002; Banta et al., 2007), GABLS3 (Bosveld et al., 2014), and SHEBA (Uttal et al., 2002; Grachev et al., 2007). Fast time-scale eigenvalue stability (λ_f) is evaluated using Geometric Singular Perturbation Theory (GSPT) analysis (Fenichel, 1979; Kuehn, 2015).

Table 1: Multi-Campaign Empirical Benchmark Synthesis and Stencil Guardrail Evaluation

Campaign	Levels (N_z)	Domain	Dominant Profile Feature	Fast Status (λ_f)	Operational Audit Result
CASES-99 (Poulos et al., 2002)	$N_z = 150$	Prairie (55m)	Severe gradient reversal ($z \approx 45\text{m}$)	Hyperbolic ($\lambda_f < -0.10$)	41.01% of profile nodes classified as curvature-dominated mapping events; 0% fast hyperbolicity loss.
GABLS3 (Bosveld et al., 2014)	$N_z = 150$	Flat Mast (200m)	Nocturnal LLJ shear erosion aloft (Sun et al., 2012)	Spurious decoupling	In tested GABLS3 SCM setup, C-Z0HR reduced nocturnal 2-m cold bias from -4.2 K (unregularized) to -1.1 K .
SHEBA (Uttal et al., 2002)	$N_z = 2$	Arctic Ice Pack	Sub-minimal stencil ($N_z < 3$)	Unresolvable $C_\eta(z, \Delta z)$	Triggers $N_z \geq 3$ stencil guardrail; safely routes to 1D Bulk Richardson fall-back.

Table 2: Low-Level Jet Profile Audit: C^0 Piecewise Ramp vs. C^∞ Regularized Ramp Closure ($Ri_c = 0.20$, $\epsilon = 10^{-3}$, $\alpha = 2.0$, $\epsilon_c = 0.05$, $\eta = 10^{-4}$, Blackadar $l_m(z)$ with $l_0 = 28.8$ m)

z (m)	$\mathcal{S}$ (s^{-1})	Ri_{raw}	$\widetilde{Ri}_g$	$\mathcal{D}_\alpha^{(R)}$	$K_{m,\text{raw}}$ (m^2/s)	$K_{m,\text{reg}}(C^0)$	$K_{m,\text{reg}}(C^\infty)$
35.88	0.07490	0.1087	0.1244	0.9558	1.43209	0.98191	0.98191
38.87	0.05561	0.1940	0.1974	0.5179	0.00510	0.00096	0.00096
40.87	0.04387	0.3087	0.2891	0.9513	0.00000	0.00000	1.50×10^{-12}
50.83	0.00343	48.544	22.733	0.5724	0.00000	0.00000	2.40×10^{-18}
60.80	0.03482	0.4553	0.4339	0.9895	0.00000	0.00000	2.77×10^{-13}

4 Conclusion

Curvature-Aware Zero-Offset Hyperbolic Regularization (C-ZOHR) provides a mathematically consistent framework for stable boundary layer parameterizations. By shifting regularization upstream to $\widetilde{Ri}_g$, binding threshold sensitivity via spatial smooth-norm curvature $C_\eta(z, \Delta z)$, and enforcing end-to-end C^∞ regularized ramp differentiability, C-ZOHR mitigates spurious quenching and Jacobian step-discontinuities while leaving legacy stability functions $f_{m,h}$ intact.

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